

Project Brief

- The Goal: Trigger DHCP, DNS and ARP protocols for basic troubleshooting.
- The Tools: CMD and WireShark
- The Method: `ipconfig /release`, `ipconfig /flushdns`, then `ipconfig /renew` to restart flow.
- Methodology: CMD to give commands, ScreenShots to document and WireShark to capture the protocols “red handed”

Setting The Conditions for the Test – part 1

- Step 1: open CMD as administrator

- Step 2: `ipconfig /release`

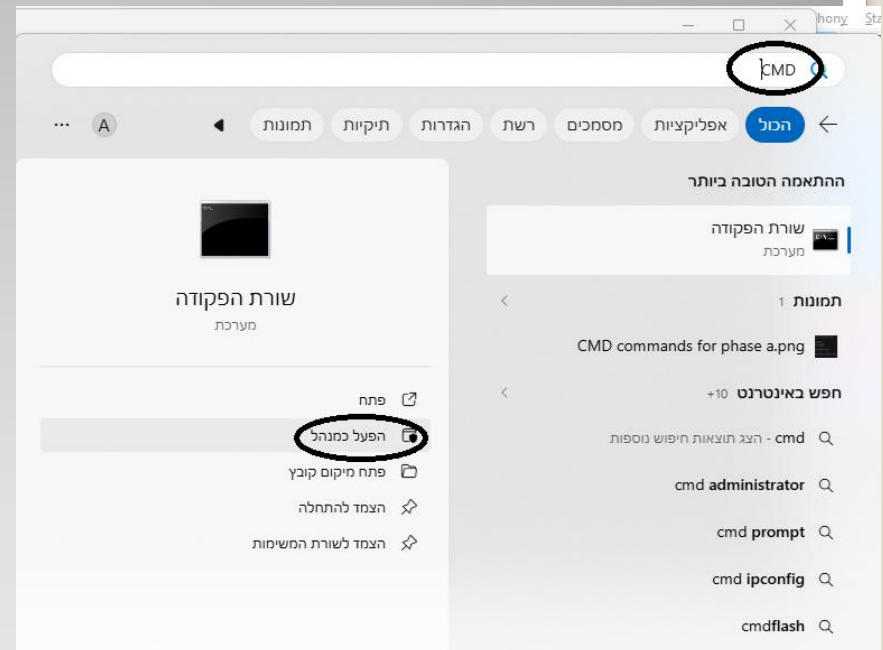
What happens now:

Your PC sends a message to the DHCP server (usually your router) saying "I'm giving back my IP address"

The IP address is released — your network adapter no longer has a valid IP

You lose internet/network connectivity immediately after running it

Your network adapter shows 0.0.0.0 as its IP address



Administrator: שורת הפקודה

```
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.
```

```
C:\Windows\System32>ipconfig /release
```



Setting The Conditions for the Test – part 2

- Step 2: ipconfig /release - DONE

- Step 3 – ipconfig /flushDNS

What happens now:

Finds the local DNS cache stored in memory
by Windows

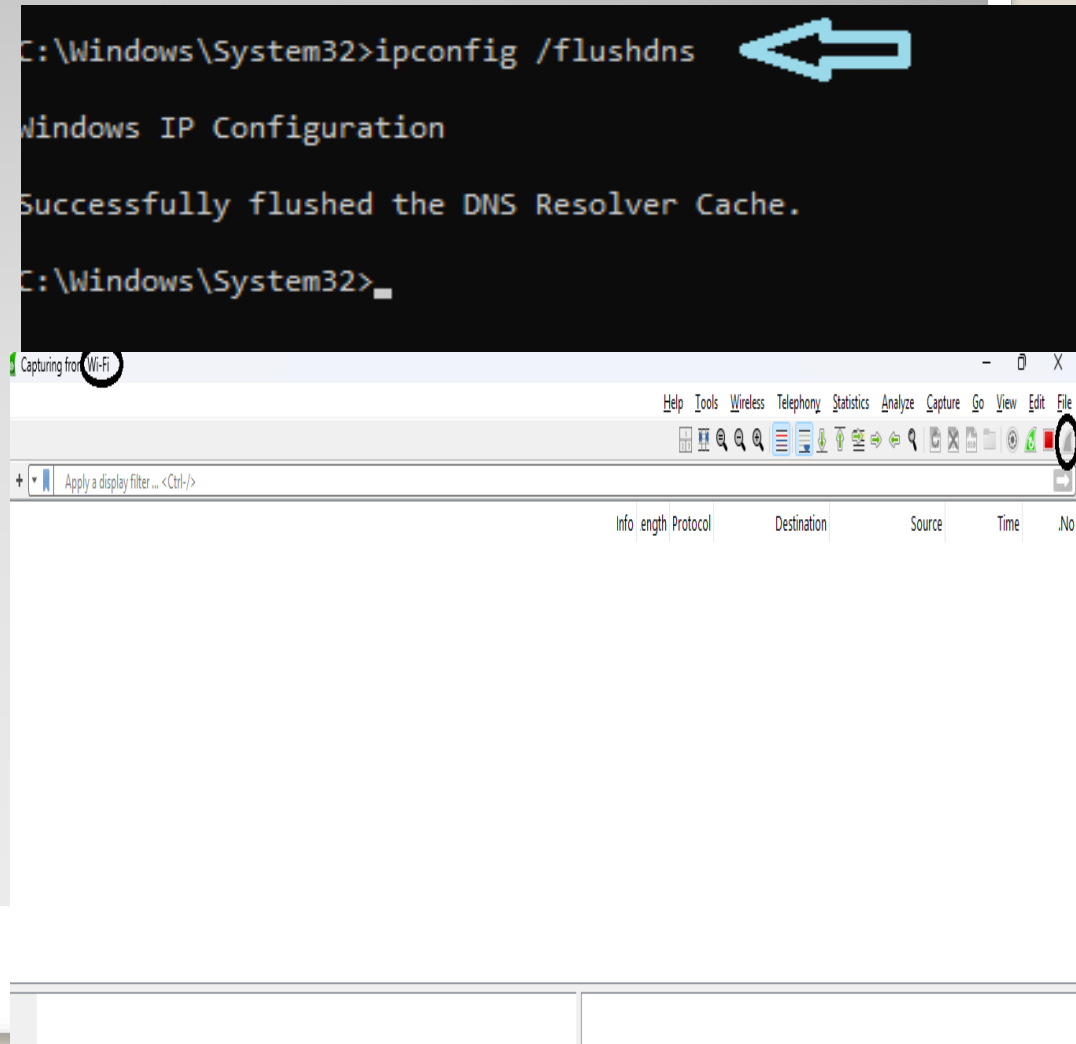
Deletes all entries from that cache

Confirms success with the message:

Successfully flushed the DNS Resolver Cache.

Next time you visit a website, your PC will do a
fresh DNS lookup instead of using cached data

- Step 4: Starting the WireShark to record
network data (from WI FI network)



Starting Test - Renewing the Cache

- Step 5: ipconfig /renew

What is happening:

The PC broadcasts a request to the DHCP server (router) saying "I need an IP address"

The router responds and assigns an IP address

The network adapter is updated with the new IP, subnet mask, default gateway, and DNS info

Network connectivity is restored

- Step 6: nslookup google.com to force data flow and DNS request

```
Administrator: שורת הפקודה
Windows IP Configuration

Successfully flushed the DNS Resolver Cache.

C:\Windows\System32>ipconfig /renew
Windows IP Configuration

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::c509:cb17:7721:a5f1%14
    IPv4 Address. . . . . : 192.168.1.5
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.1

Ethernet adapter לש תשר רוביח Bluetooth:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

C:\Windows\System32>nslookup google.com
Server: InnboxG84
Address: 192.168.1.1

Non-authoritative answer:
Name: google.com
Addresses: 2a00:1450:4028:80a::200e
          142.250.75.174
```

IP Config /all – See current connections

- The command:

Ipconfig /all – returns current connections.

Looking at the WIFI LAN we see this:

- Physical address (MAC number): 30-D1-

5B-F3-09-F1

- Preferred PC IP: **192.168.1.5**

- Default Gateway: **192.168.1.1**

- The Default Gateway (router) is also the

DHCP and DNS server

```
C:\Users\97252>ipconfig /all
```

```
Adapter #4
Physical Address. . . . . : 42-D1-6B-F3-09-F1
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Ethernet 2:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 
Description . . . . . : TAP-Windows Adapter V9
Physical Address. . . . . : 00-FF-58-7D-0C-D6
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . : 
Description . . . . . : Qualcomm Atheros QCA9377 Wireless
Network Adapter
Physical Address. . . . . : 30-D1-6B-F3-09-F1
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::c509:cb17:7721:a5f1%14(Preferred)
IPv4 Address. . . . . : 192.168.1.5(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : יום יום 19 2026 17:20:31
Lease Expires . . . . . : יום יום 20 2026 17:26:39
Default Gateway . . . . . : 192.168.1.1
DHCP Server . . . . . : 192.168.1.1
DHCPv6 IAID . . . . . : 134194285
DHCPv6 Client DUID. . . . . : 00-01-00-01-23-64-68-9A-E8-6A-64
DNS Servers . . . . . : 192.168.1.1
NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Bluetooth:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
```

Results – WireShark ARP

What is ARP? ARP (Address Resolution Protocol) is how devices on a network translate an **IP address** → **MAC address**. Before any communication can happen on a local network, a device must know the MAC address of the target IP.

- **2nd line:** Who has 192.168.1.5? Tell 192.168.1.1 -> The router wants to ensure the IP address is available to be assigned to the PC (192.168.1.5)

Source: LiteonTechno_f3:09 → **this is the PC** (MAC 30:D1:6B:F3:09:F1)

The **router (192.168.1.1)** is asking *"who has the IP I just assigned?"* - verifying no conflict

- **3rd is at 30:d1:6b:f3:09:f1 → 192.168.1.5**

The PC replies: *"That's me! My MAC is 30:D1:6B:F3:09:F1"*

arp							
	Info	Length	Protocol	Destination	Source	Time	.No
	ARP Announcement for 192.168.1.10	46	ARP	Broadcast ...:XiaomiMobile_86:87		3.145413000	15
	Who has 192.168.1.5? Tell 192.168.1.1	42	ARP	...:LiteonTechno_f3:09	IskratelDoo_21:1e:50	33.089535500	576
	is at 30:d1:6b:f3:09:f1 192.168.1.5	42	ARP	IskratelDoo_21:1e:50	...:LiteonTechno_f3:09	33.089594900	577
	ARP Announcement for 192.168.1.10	46	ARP	Broadcast ...:XiaomiMobile_86:87		63.152309100	902
	ARP Announcement for 192.168.1.10	46	ARP	Broadcast ...:XiaomiMobile_86:87		123.159276900	1167
	Who has 192.168.1.5? Tell 192.168.1.1	42	ARP	...:LiteonTechno_f3:09	IskratelDoo_21:1e:50	176.090356400	1349
	is at 30:d1:6b:f3:09:f1 192.168.1.5	42	ARP	IskratelDoo_21:1e:50	...:LiteonTechno_f3:09	176.090414900	1350
	ARP Announcement for 192.168.1.10	46	ARP	Broadcast ...:XiaomiMobile_86:87		183.166259700	1363

Results – WireShark DNS – Part 1

- The DNS filter yields too many results.

Why So Many DNS Lines?

When running **nslookup** google.com, Windows also had **background processes** firing DNS queries simultaneously (Teams, Windows Update, Google services, etc.).

Who triggered it	Domain
Google background service	clientservices.googleapis.com
Microsoft Teams running in background	config.teams.microsoft.com
Windows telemetry	settings-win.data.microsoft.com
Google SSL service	ssl.gstatic.com
Microsoft Office background	exo.nel.measure.office.net

*Wi-Fi

Help Tools Wireless Telephony Statistics Analyze Capture Go View Edit File

+ dns

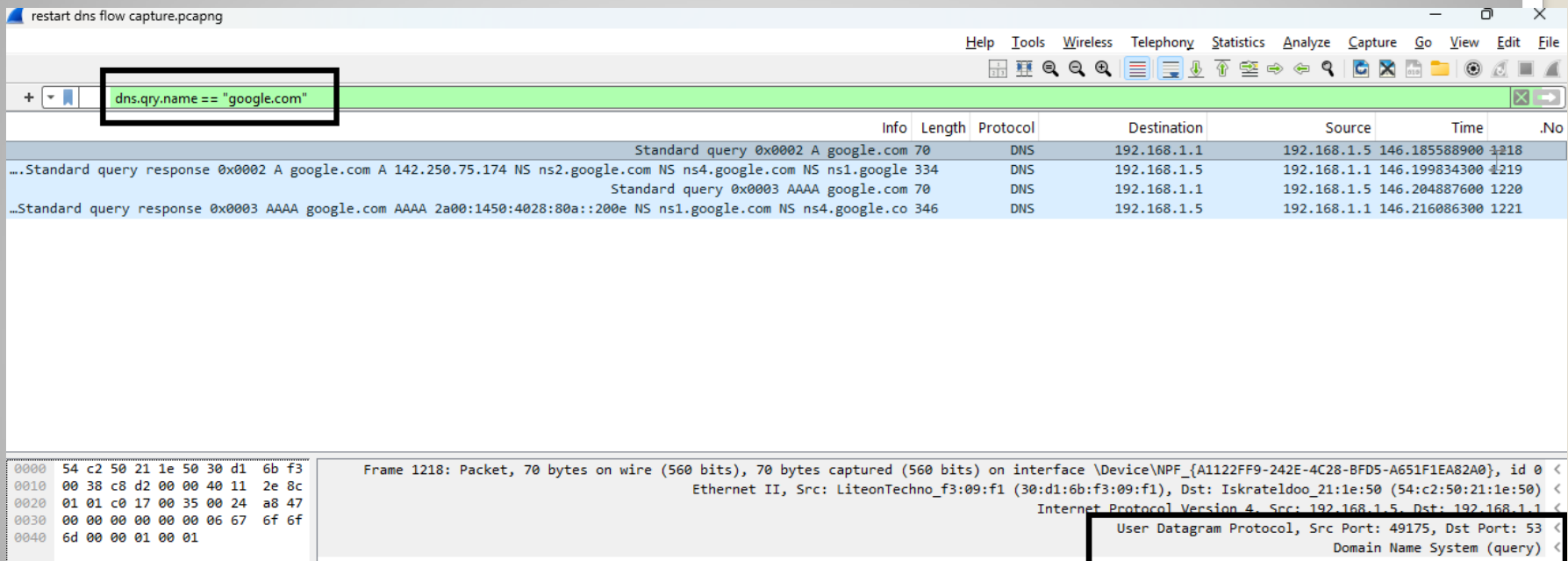
	Info	length	Protocol	Destination	Source	Time	.No
	Standard query 0x11a1 HTTPS clientservices.googleapis.com	89	DNS	192.168.1.1	192.168.1.5	19.631881500	34
	Standard query 0x1554 A clientservices.googleapis.com	89	DNS	192.168.1.1	192.168.1.5	19.632432700	35
	Standard query response 0x11a1 HTTPS clientservices.googleapis.com SOA ns1.google.com	146	DNS	192.168.1.5	192.168.1.1	19.654961300	36
...	Standard query response 0x1554 A clientservices.googleapis.com A 142.250.75.174 NS ns3.google.com NS ns2.g	360	DNS	192.168.1.5	192.168.1.1	19.655141200	37
	Standard query 0x6f4f HTTPS config.teams.microsoft.com	86	DNS	192.168.1.1	192.168.1.5	21.602395700	61
	Standard query 0xb8b3 A config.teams.microsoft.com	86	DNS	192.168.1.1	192.168.1.5	21.603006500	62
...	Standard query response 0x6f4f HTTPS config.teams.microsoft.com CNAME config.teams.trafficmanager.net CNAM	342	DNS	192.168.1.5	192.168.1.1	21.616932500	63
...	Standard query response 0xb8b3 A config.teams.microsoft.com CNAME config.teams.trafficmanager.net CNAME mi	469	DNS	192.168.1.5	192.168.1.1	21.621083100	64
	Standard query 0xe830 A settings-win.data.microsoft.com	91	DNS	192.168.1.1	192.168.1.5	22.170606800	110
...	Standard query response 0xe830 A settings-win.data.microsoft.com CNAME atm-settingsfe-prod-geo2.trafficman	540	DNS	192.168.1.5	192.168.1.1	22.184894200	112
	Standard query 0x80b7 HTTPS ssl.gstatic.com	75	DNS	192.168.1.1	192.168.1.5	22.208094700	114
	Standard query 0xc854 A ssl.gstatic.com	75	DNS	192.168.1.1	192.168.1.5	22.208705800	115
	Standard query response 0x80b7 HTTPS ssl.gstatic.com SOA ns1.google.com	132	DNS	192.168.1.5	192.168.1.1	22.225680700	119
...	Standard query response 0xc854 A ssl.gstatic.com A 142.250.75.99 NS ns3.google.com NS ns4.google.com NS ns	346	DNS	192.168.1.5	192.168.1.1	22.225680700	120
	Standard query 0xab57 HTTPS exo.nel.measure.office.net	86	DNS	192.168.1.1	192.168.1.5	22.574761800	348
	Standard query 0x405e A exo.nel.measure.office.net	86	DNS	192.168.1.1	192.168.1.5	22.575490300	349

0000 54 c2 50 21 1e 50 30 d1 6b f3 09 f1 08 00 45 00 T.P!·P0· k·...·E·
0010 00 4b c8 b8 00 00 40 11 2e 93 c0 a8 01 05 c0 a8 ·K·...@· ·...
0020 01 01 ff d3 00 35 00 37 9f 93 11 a1 01 00 00 01 ·...·5·7 ·...
0030 00 00 00 00 00 00 0e 63 6c 69 65 6e 74 73 65 72 ·...·c lientser
0040 76 69 63 65 73 0a 67 6f 6f 67 6c 65 61 70 69 73 vices·go ogleapis
0050 03 63 6f 6d 00 00 41 00 01 ·com··A· ·

ptured (712 bits) on interface \Device\NPF_{A1122FF9-242E-4C28-BFD5-A651F1EA82A0}, id 0 <
iteonTechno_f3:09:f1 (30:d1:6b:f3:09:f1), Dst: IskratelDoo_21:1e:50 (54:c2:50:21:1e:50) <
Internet Protocol Version 4, Src: 192.168.1.5, Dst: 192.168.1.1 <
User Datagram Protocol, Src Port: 65491, Dst Port: 53 <
Domain Name System (query) <

DNS Results – WireShark– Part 2

- After adjusting the filter from DNS to `dns.qry.name == "google.com"` we receive the DNS triggered by the `nslookup` command
- Here you can see 2x2 packets of “query” and “response” (PC, router)
- In the Query data we can see **UDP** protocol and **port 53** – meaning it’s DNS (always)



The image shows a Wireshark capture window titled "restart dns flow capture.pcapng". The filter bar at the top contains the expression `dns.qry.name == "google.com"`, which is highlighted with a black box. Below the filter bar, a table of captured packets is displayed. The table has columns for "Info", "Length", "Protocol", "Destination", "Source", "Time", and ".No". The packets are as follows:

Info	Length	Protocol	Destination	Source	Time	.No
Standard query 0x0002 A google.com 70	70	DNS	192.168.1.1	192.168.1.5	146.185588900	1218
...Standard query response 0x0002 A google.com A 142.250.75.174 NS ns2.google.com NS ns4.google.com NS ns1.google.com 334	334	DNS	192.168.1.5	192.168.1.1	146.199834300	1219
Standard query 0x0003 AAAA google.com 70	70	DNS	192.168.1.1	192.168.1.5	146.204887600	1220
...Standard query response 0x0003 AAAA google.com AAAA 2a00:1450:4028:80a::200e NS ns1.google.com NS ns4.google.co 346	346	DNS	192.168.1.5	192.168.1.1	146.216086300	1221

Below the packet list, the details pane for "Frame 1218: Packet, 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface \Device\NPF_{A1122FF9-242E-4C28-BFD5-A651F1EA82A0}, id 0" is shown. The protocol stack is expanded to show "User Datagram Protocol, Src Port: 49175, Dst Port: 53", which is highlighted with a black box. The "Domain Name System (query)" section is also visible below it.

Packet bytes (hex):

```
0000 54 c2 50 21 1e 50 30 d1 6b f3
0010 00 38 c8 d2 00 00 40 11 2e 8c
0020 01 01 c0 17 00 35 00 24 a8 47
0030 00 00 00 00 00 06 67 6f 6f
0040 6d 00 00 01 00 01
```


DHCP Results – WireShark– Part 1

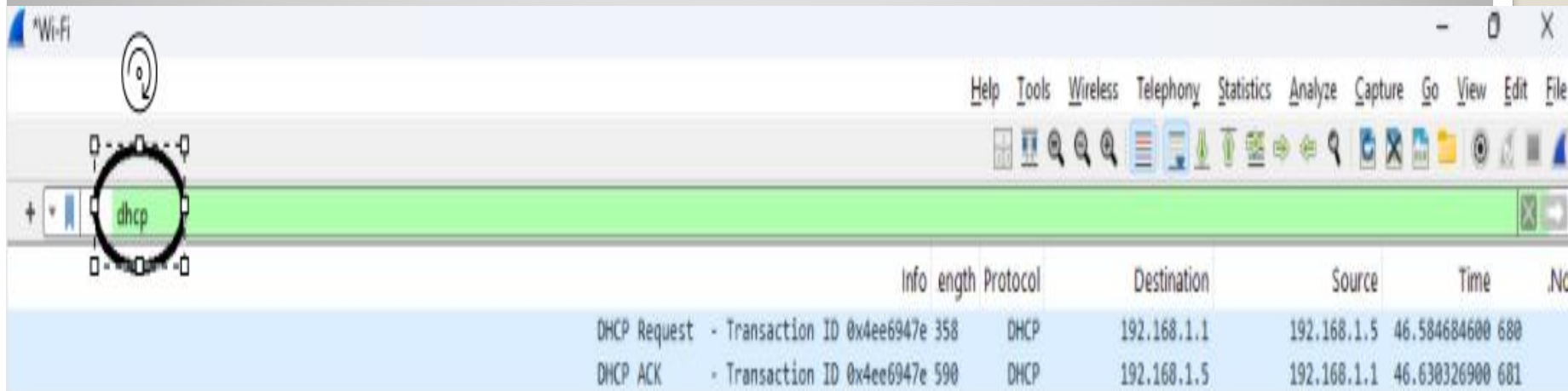
- After filtering the WireShark recording to DHCP we can see 2 DHCP commands.

- 1st the PC requests an IP from the router (the DHCP server).

- 2nd the router acknowledges the DHCP request and returns to the client

192.168.1.5 (the PC) -> DHCP request -> 192.168.1.1 (the router) -> DHCP ACK (response from the router)

- **This result shows only 2 of the 4 steps of DORA, because the PC remembered it's preferred IP and asked for it immediately when the process started (command **renew**)**



The screenshot shows the Wireshark network protocol analyzer interface. The top menu bar includes Help, Tools, Wireless, Telephony, Statistics, Analyze, Capture, Go, View, Edit, and File. The toolbar contains various icons for packet analysis. The packet list pane shows two packets:

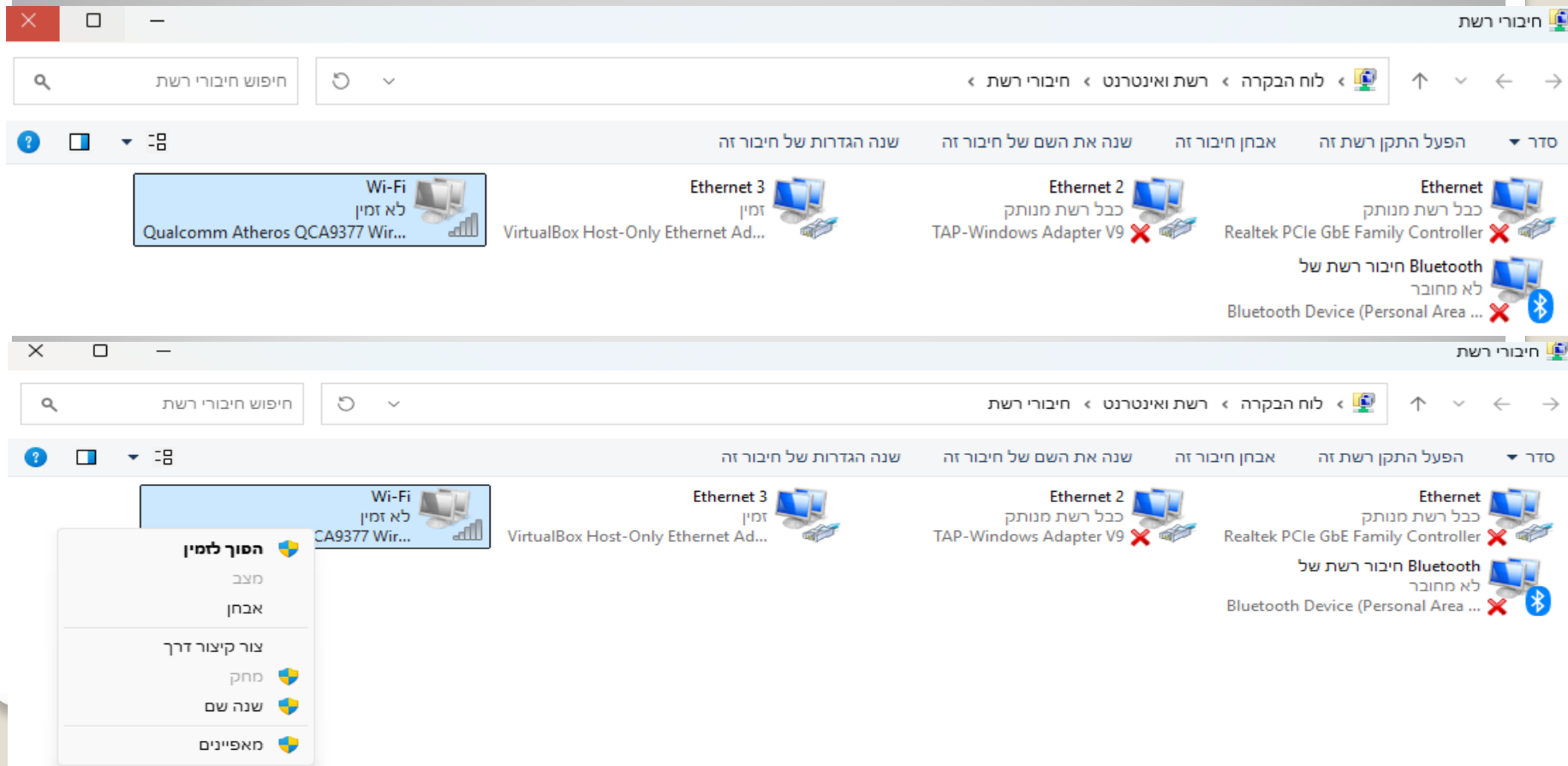
No.	Time	Source	Destination	Protocol	Length	Info
1	46.584684600	192.168.1.5	192.168.1.1	DHCP	358	DHCP Request - Transaction ID 0x4ee6947e
2	46.630326900	192.168.1.1	192.168.1.5	DHCP	590	DHCP ACK - Transaction ID 0x4ee6947e

The packet details pane shows the selected packet (DHCP Request) with the following structure:

- Frame 1: 358 bytes on wire (2864 bits) captured (2864 bits) on interface 0
- Ethernet II, Src: Realtek-USB (82:55:88:53:89:00), Dst: 01:00:00:00:00:00
- Internet Protocol Version 4, Src: 192.168.1.5, Dst: 192.168.1.1
- DHCP, Transaction ID: 0x4ee6947e

DISABLE/ENABLE Network To Force Fresh DHCP

In order to see the DHCP refreshes and see the 4 DORA steps of DHCP we need to disable the WIFI network and restart it



DHCP Results – WireShark– Part 2

- Now we can see the 4 stages of **DORA** when asking for a new IP
- **1st DISCOVER** - the PC has no IP and doesn't know the router (or anyone in the network). That's why the source (the PC) IP is 0.0.0.0 and the destination is 255.255.255.255 (requesting DHCP from anyone listening)
- **2nd OFFER** – the router picks up the DHCP call and offers **192.168.1.5** as IP.
- **3rd REQUEST** – the PC request the IP address
- **4th ACK** – the router acknowledges the new IP distributed to the PC

Capturing from Wi-Fi

Help Tools Wireless Telephony Statistics Analyze Capture Go View Edit File

dhcp

Enter a filter expression to be applied Filter: Enter a description for the filter button Label: ...Filter Buttons Preferences

ביטול אישור

Enter a comment for the filter button Comment:

	Info	Length	Protocol	Destination	Source	Time	.No
	DHCP Discover - Transaction ID 0x1728976c	299	DHCP	255.255.255.255	0.0.0.0	0.004706600	3
	DHCP Offer - Transaction ID 0x1728976c	590	DHCP	192.168.1.5	192.168.1.1	0.168159300	13
	DHCP Request - Transaction ID 0x1e14e61a	364	DHCP	255.255.255.255	0.0.0.0	0.444760300	33
	DHCP ACK - Transaction ID 0x1e14e61a	590	DHCP	192.168.1.5	192.168.1.1	0.487595500	34

THANK YOU FOR WATCHING